

Keywords: RLVR • entropy reward • LLM reasoning efficiency • chain-of-thought

ERR+: Sequential Entropy Resolution

Rewarding Token-Level Entropy Drops Teaches Decisive Reasoning and Cuts Chain-of-Thought Length on AIME, AMC, and STEM Benchmarks

By Juxin Chen et al.

arXiv:2608.28771 • Findings of EMNLP 2026

Correct chain-of-thought traces collapse token-level entropy more frequently and more sharply than incorrect ones. ERR+ rewards these drops in Phase 1 and compresses trace length in Phase 2, improving accuracy by 2–5 points and reducing trace length by 15–25% on five math and STEM benchmarks.

AT A GLANCE

Problem: RLVR-trained large reasoning models produce accurate but verbose traces; existing length penalties cut indiscriminately and degrade accuracy.

Why it matters: Long traces waste inference compute; for deployed LRMs on math and STEM tasks, shorter decisive reasoning directly reduces serving cost.

Method: Two-phase RLVR: Phase 1 rewards log-normalized cumulative entropy drops (gated on correctness); Phase 2 compresses length relative to rollout-group peers via GRPO.

Result: +2–5 pp accuracy, –15–25% trace length on AIME, AMC, MATH-500, GPQA-Diamond, STEM multi-step.

Evidence: Ablations across 4 design axes confirm both phases are necessary; log normalization critical for preventing length gaming.

The Big Picture

Reinforcement learning with verifiable rewards (RLVR) has produced state-of-the-art reasoning models. These models generate extended chain-of-thought (CoT) traces—sometimes spanning thousands of tokens—before emitting a final answer. The length is partly necessary: complex multi-step problems genuinely require extended computation. But a substantial fraction of trace tokens represents redundant backtracking, repeated reformulation, and exploratory text that does not advance the reasoning.

The tension is delicate. Simply penalising length reduces redundancy but also cuts useful deliberation, leading to accuracy regression. What is needed is a way to distinguish

decisive reasoning steps—moments when the model resolves uncertainty and commits to a path—from *indecisive* ones that wander or repeat.

ERR+ proposes that token-level entropy drops are precisely this signal. When the model collapses its next-token distribution sharply, it has made a committed inference. Correct reasoning chains do this more frequently than incorrect ones, and rewarding this pattern produces models that are simultaneously more accurate *and* more concise.

Why Existing Approaches Fall Short

Binary outcome RLVR rewards all correct traces equally, regardless of whether they are decisive or verbose. It provides no gradient signal toward more efficient reasoning.

Length penalties (absolute or relative) reduce trace length but penalise useful long reasoning steps alongside useless ones. The accuracy degradation is well-documented.

KL regularisation toward a reference model prevents entropy collapse globally but does not specifically encourage entropy drops *during* the thinking phase—it regulates the overall distribution, not the dynamics within a trace.

Min-token objectives force early answer commitment, degrading accuracy on hard problems that genuinely require extended reasoning.

ERR+ is the first method to use the temporal *pattern* of entropy changes within the thinking phase as a training signal, rather than aggregate trace length or final-state entropy.

Core Mechanism

The Entropy Drop Observation. Let $H_t = -\sum_v p_\theta(v|x_{<t}) \log p_\theta(v|x_{<t})$ be the next-token entropy at position t within the thinking phase $\mathcal{T}_{\text{think}}$. An entropy drop at position t is $\Delta H_t = H_{t-1} - H_t > 0$.

Empirical analysis across multiple model families (7B–70B) reveals:

- Correct reasoning traces have 30–50% more entropy drops on average than incorrect ones on MATH and AIME tasks
- The magnitude of drops is also larger in correct traces
- This pattern holds even when controlling for total trace length

Phase 1: Entropy Relief Reward (ERR). The cumulative entropy drop, log-normalized and gated on correctness, forms the ERR bonus:

$$\text{ERR}(\tau) = \frac{\sum_{t \in \mathcal{T}_{\text{think}}} \max(0, H_{t-1} - H_t)}{\log(|\tau| + 1)} \cdot \mathbf{1}[a(\tau) = a^*]$$

Phase 2: Group-Relative Length Compression. A GRPO-style penalty rewards traces that are shorter than their rollout-group average:

$$r_{\text{len}}(\tau) = -\gamma \cdot \frac{|\tau| - |\bar{\tau}|_{\text{group}}}{|\bar{\tau}|_{\text{group}}}$$

Technical Formulation

Token-level entropy:

$$H_t = - \sum_{v \in \mathcal{V}} p_{\theta}(v \mid x_{<t}) \log p_{\theta}(v \mid x_{<t})$$

Log-normalized entropy relief:

$$\text{ERR}(\tau) = \frac{1}{\log(|\tau| + 1)} \sum_{t \in \mathcal{T}_{\text{think}}} \max(0, H_{t-1} - H_t) \cdot \mathbf{1}[a(\tau) = a^*]$$

Phase 1 total reward:

$$r_1(\tau) = r_{\text{base}}(\tau) + \beta \cdot \text{ERR}(\tau)$$

Phase 2 total reward:

$$r_2(\tau) = r_{\text{base}}(\tau) - \gamma \cdot \frac{|\tau| - |\bar{\tau}|_{\text{group}}}{|\bar{\tau}|_{\text{group}}}$$

Phase 1 policy gradient:

$$\nabla_{\theta} \mathcal{L}_1 = -\mathbb{E}_{\tau}[r_1(\tau) \cdot \nabla_{\theta} \log p_{\theta}(\tau)]$$

Learning or Inference Procedure

1. Start from a CoT-fine-tuned base model.
2. **Phase 1:** Run RLVR with r_1 for N_1 steps; monitor accuracy and mean CED (cumulative entropy drop).
3. **Phase 2:** Continue from Phase 1 checkpoint with r_2 for N_2 steps; monitor accuracy and trace length.
4. Evaluate at fixed inference budget (greedy, no search).

What the Guarantee Says

The log-normalization in ERR provides a formal argument against length gaming: the ERR bonus grows only as $O(\log |\tau|)$ with length, while the number of possible entropy drops grows as $O(|\tau|)$. A model that increases length purely to accumulate entropy drops earns a $O(|\tau|/\log |\tau|)$ bonus but pays increasing KL cost—net loss for long random traces.

Experimental Findings

- **Benchmarks:** AIME 2024, AMC 2023, MATH-500, GPQA-Diamond, custom STEM multi-step dataset

- Accuracy improves by 2–5 pp over RLVR-only baseline
- Trace length decreases by 15–25% relative to Phase 1 baseline
- ERR+ outperforms length-penalty-only methods at matched length
- Mean CED increases by $\approx 40\%$ after Phase 1 training, confirming decisiveness is learned, not length

Ablations and Interpretation

Phase 1 only: Accuracy improves; trace length unchanged. ERR alone drives decisiveness but not compression.

Phase 2 only (no Phase 1): Length decreases; accuracy *degrades* by 3–4 pp. Phase 1 first establishes the decisive reasoning style that Phase 2 then compresses.

No log-normalization: Accuracy improves; trace length *increases*—the model games the reward by generating more drops absolutely, confirming that normalization is essential.

No correctness gating: Entropy drops in incorrect traces are also rewarded; reasoning quality degrades, confirming that decisive wrong reasoning is worse than exploratory wrong reasoning.

β sweep: Optimal $\beta \in [0.2, 0.5]$; large β collapses entropy across all positions, hurting exploration.

Reference

Juxin Chen et al. **ERR+: Sequential Entropy Resolution for Efficient and Decisive LLM Reasoning.** arXiv:2608.28771. Findings of EMNLP 2026.
<https://arxiv.org/abs/2608.28771>